04 · Which funnel stage to invest in? — mediation (pathmc)

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The business decision. We have a fixed budget to split across the growth funnel — **onboarding → engagement → activation → conversion** — and want to put it where it moves paid conversion the most. The complication is that the stages form a **chain**: money spent improving onboarding doesn't only help onboarding, it *cascades* downstream into engagement, activation, and ultimately conversion. So we can't just correlate each stage with revenue; we have to trace the causal *paths* and decide where a push travels furthest.

The concept: mediation, direct and indirect effects

When a treatment X affects an outcome Y *through* an intermediate variable M (a **mediator**), the total effect splits in two:

- the **direct effect** — X's influence on Y that does *not* go through M (Pearl's *natural direct effect*, NDE);
- the **indirect effect** — the part that flows $X \to M \to Y$ (the *natural indirect effect*, NIE).

Total effect = direct + indirect, and the **proportion mediated** = indirect / total tells you how much of a stage's value is really realised *downstream*. In a linear model the indirect effect is just the **product of the coefficients along the path** (the classic "a·b" of mediation analysis) — `pathmc` gives us exactly this via `effect(path)`, which returns the **posterior product-of-coefficients along the path** (exact for a linear SCM — not a Monte-Carlo simulation through the graph), so we get every path's effect *with uncertainty*.

The catch this notebook is honest about

Mediation leans on a **stronger** assumption than a simple treatment-effect analysis: no unmeasured confounding of the *mediator–outcome* link (on top of the usual treatment–outcome one). If some unmeasured trait drives both engagement *and* conversion, the direct/indirect *split* is biased even when the *total* effect is fine. So Step 6 doesn't just report the split — it **stress-tests how much such confounding would have to exist to flip the "invest here" decision.**

Why not just A/B test each stage?

Often you can — and for the **total** effect of one lever, you should. But two things stop experiments from answering the funnel question on their own. First, mid-funnel stages are hard to randomize *directly*: you can assign a user a new onboarding flow, but you cannot assign them an engagement level — engagement is a behaviour, not a knob. Second, even a perfectly randomized onboarding test identifies only the **total** effect; the direct/indirect *split* — "does onboarding pay off by itself, or only by feeding engagement?" — still needs the mediator-side assumptions of Step 3. That is the honest positioning of mediation analysis: it answers a question experiments alone cannot ("*where* *in the chain* does the value travel?"), at the price of stronger assumptions — which is why this notebook spends so much time naming, testing, and stress-testing them.

> **On real data.** Swap in your **product-analytics event data** — per-user onboarding score, engagement, activation flag, conversion — which every growth team already logs. The method needs the funnel's causal *order* to be known (it usually is) and the no-hidden-mediator-confounder assumption to be defensible (Step 6 shows how to argue it).

`pathmc` builds the chain as a structural causal model and computes **path-specific effects** as the posterior **product-of-coefficients along each path** — exact for a linear SCM (we reserve "simulate through the intervened graph" for `do()`/`ate()`) — the modern, uncertainty-aware mediation analysis.

It follows the repo's fixed **7-step contract**: question → simulate → identify → estimate → validate → decide in € → caveats.

```
FAST=False  sampling={'draws': 1500, 'tune': 1500, 'chains': 4}
```

2 · Simulate a ground truth

We simulate the funnel as a **causal chain**: `onboarding_score` → `engagement` → `activated` → `converted`, with `channel_quality` feeding both engagement and activation (a common driver we'll need to account for). Each arrow is a real structural equation with a planted coefficient, so we know the **true** total effect of onboarding on conversion *and how it splits across paths* — which is exactly what we'll ask the model to recover. By construction most of onboarding's effect flows the long way round (`engagement` → `activated` → `converted`), with a smaller slice going `engagement` → `converted` directly. If the method can rediscover that split from data alone, we can trust it on a funnel where we *don't* know the answer.

The data-generating model — exactly what `dgp.funnel` implements (defaults & seed in `src/cmp/dgp.py`). Onboarding $O \sim U(0,1)$ (uniform on $[0,1]$ — equally likely anywhere in between) and channel quality $Q \sim U(0,1)$ are **exogenous** — determined outside the model, with no arrow pointing into them (both are simply drawn at random); each downstream stage is a linear structural equation:

$$\begin{aligned} \text{engagement} &= 2.2\,O + 1.1\,Q + \varepsilon_1, & \varepsilon_1 &\sim \mathcal{N}(0,\,0.4^2)\\ \text{activated} &= 1.6\,\text{engagement} + 0.6\,Q + \varepsilon_2, & \varepsilon_2 &\sim \mathcal{N}(0,\,0.4^2)\\ \text{converted} &= 0.9\,\text{activated} + 0.35\,\text{engagement} + \varepsilon_3, & \varepsilon_3 &\sim \mathcal{N}(0,\,0.3^2). \end{aligned}$$

The planted path effects are the **products of coefficients along each route** — exactly what the mediation analysis must recover: the long path $O \to \text{eng} \to \text{act} \to \text{conv} = 2.2 \times 1.6 \times 0.9 = 3.168$, the short path $O \to \text{eng} \to \text{conv} = 2.2 \times 0.35 = 0.77$, and **no** direct $O \to \text{conv}$ arrow (true NDE $= 0$), so the true total effect is 3.938 and the true proportion mediated is 100%. Q feeds two stages (engagement *and* activation) — the common driver the model must include to keep the paths clean.

```
TRUE effects of onboarding on conversion:
  indirect_via_engagement_activated    3.17
  indirect_via_engagement_only         0.77
  total                                3.94
```

	onboarding_score	channel_quality	engagement	activated	converted
0	0.864798	0.726815	2.576471	3.869429	4.870321
1	0.855303	0.790067	3.206651	5.673260	6.371102
2	0.811023	0.348493	2.493785	4.691864	5.394113
3	0.261446	0.465793	0.572682	1.261547	1.264003
4	0.077199	0.887974	1.520775	3.259916	3.146544

3 · Identify — direct vs indirect effects (NDE / NIE)

For a chain $X \to M \to Y$ the **total effect** decomposes into a **natural direct effect** and **natural indirect effect** (Pearl):

$$\text{TE} = \text{NDE} + \text{NIE}, \quad \text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \text{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Here $Y(x, m)$ is the conversion we would see if onboarding were set to x **and** the mediator to m, and $M(x)$ is the mediator value onboarding x would produce — so the NDE holds the mediator at its *untreated* natural value and moves only the treatment, while the NIE holds the treatment fixed and swaps which world the mediator comes from. The **proportion mediated** = NIE/TE tells you how much of onboarding's value is "really" downstream engagement/activation. Crucially we **include a direct onboarding→converted edge in the model and test it**, so a near-100%-mediated answer is *discovered* (the coefficient comes out ≈ 0), not forced by leaving the edge out.

What buys the decomposition — sequential ignorability, one condition at a time

The NDE/NIE are **cross-world** quantities — $Y(1, M(0))$ mixes a treated outcome with an untreated mediator, a combination *no experiment ever shows you*. Identifying them takes four conditions (from the mediation-analysis literature — Imai et al.; VanderWeele), stated with X = onboarding, M = the mediator(s), Y = conversion, and C = the observed common drivers (here `channel_quality`):

Reading the notation. $A \perp B \mid C$ means "once we know C, A carries no information about B" (they are statistically independent given C); $\mathbb{E}[\cdot]$ is an average; $\int \cdots dP(\cdot)$ is that same average written as an integral — a probability-weighted sum over the values of whatever follows dP; and a **prime** (x') marks a *possibly different* treatment value than x — that mismatch between the world the outcome lives in and the world the mediator comes from is exactly what makes these quantities "cross-world."

Ignorability itself means: given the controls C, who got the treatment is as-good-as-random, so we can "ignore" how assignment happened. Ordinary treatment-effect analysis needs this once (SI-1); mediation needs it **again for the mediator** (SI-2, SI-4) — that extra demand is why it is called *sequential* ignorability and why it is strictly stronger.

$$\text{(SI-1)} \qquad \{\, Y(x, m),\ M(x')\,\} \ \perp\ X \mid C$$

No unmeasured treatment-side confounding: given C, who got good onboarding is as-good-as-random with respect to both outcome and mediator. **Violation:** highly motivated users both breeze through onboarding *and* would convert anyway — motivation is unlogged, and onboarding inherits its credit.

$$\text{(SI-2)} \qquad Y(x,m) \;\perp\; M \;\big|\; X{=}x,\; C$$

No unmeasured mediator–outcome confounding — **the killer**, because M is a behaviour, never randomized even inside an RCT. **Violation:** a user's intrinsic interest in the product drives both their engagement *and* their conversion; the fitted engagement→conversion edge then overstates the mechanism and the direct/indirect split is wrong even when the total is fine. This is the assumption Step 6 stress-tests quantitatively.

$$\text{(SI-3)} \qquad 0 \;<\; p(M{=}m \mid X,\, C) \;\; \text{for all } m \;\; \text{(positivity)}$$

Every mediator level must be *reachable* at every treatment level — if no low-onboarding user ever shows high engagement, the counterfactual "high engagement without onboarding" is pure extrapolation.

$$\text{(SI-4)} \qquad Y(x,m) \;\perp\; M(x') \;\big|\; C \qquad \text{(cross-world independence)}$$

No mediator–outcome confounder that is *itself caused by* X — e.g. onboarding must not create a habit that separately drives both engagement and conversion. This one is **untestable even in principle** (the two sides live in different worlds), which is the honest price of asking a cross-world question.

The mediation formula — what the assumptions buy

Under SI-1–SI-4, the cross-world mean is identified by the **mediation formula** (Pearl/Imai):

$$\mathbb{E}\big[Y(x,\, M(x'))\big] \;=\; \int_m \mathbb{E}\big[\, Y \mid X{=}x,\; M{=}m,\; C \,\big]\, dP(m \mid X{=}x',\; C)\, dP(C).$$

Read it as a recipe — the outer $\int \cdots\, dP(m \mid X{=}x', C)$ is exactly the "average over the mediator values that treatment x' would produce" from the notation key above: predict the outcome under treatment x, but average over the mediator distribution that treatment x' *would have produced.* This is the estimator-agnostic result — everything `pathmc` computes below is this formula specialized to a linear Gaussian (Gaussian = normal) SCM.

From formula to "product of coefficients" — two lines

For the simple chain with one mediator, let $M = a\,X + \varepsilon_M$ and $Y = c\,X + b\,M + \varepsilon_Y$ (noise mean-zero, no $X{\times}M$ interaction). Then $\mathbb{E}[Y \mid X{=}x, M{=}m] = c\,x + b\,m$ and $\mathbb{E}[M \mid X{=}x'] = a\,x'$, so the mediation formula collapses to

$$\mathbb{E}\big[Y(x, M(x'))\big] \;=\; c\,x \;+\; b\,a\,x' \qquad \Longrightarrow \qquad \text{NDE} = c\,\Delta x, \quad \text{NIE} = a\,b\,\Delta x, \quad \text{TE} = (c + ab)\,\Delta x.$$

That is the classic $a \cdot b$ of mediation analysis, now *derived* rather than asserted — and it is why `effect(path)` returns the **posterior product of the coefficients along the path**, exact for a linear SCM (not a Monte-Carlo simulation through the graph), so every path effect comes *with uncertainty*. One subtlety for our **continuous** treatment: the effects here are per-unit contrasts ($x \to x+1$), and because the SCM is linear with no interaction they are the same at every x — with nonlinearity or interaction, NDE/NIE would depend on the baseline and the clean additivity above would need the four-way (VanderWeele) decomposition (Step 7).

Two mediators in sequence — path-specific effects need one more condition

Our funnel has **two ordered mediators** (engagement → activated), so "indirect via activated" is a *path-specific effect*, and its identification needs, on top of SI-1–SI-4, **no unmeasured confounding between the mediators themselves**. That is exactly why the DGP's `channel_quality` — which feeds *both* engagement and activation — must be observed and adjusted: leave it out and the engagement→activated edge silently absorbs its effect, poisoning every path product built on that edge. Step 5 doesn't leave this as a warning: we *refit without it and watch the split break.*

Which assumption breaks first in practice? SI-2 (and its mediator–mediator cousin) — you can randomize onboarding but never engagement. So Step 6 doesn't just report the split; it measures how much hidden mediator–outcome confounding it would take to change the *decision.*

4 · Estimate — fit the chain and query every path

We hand `pathmc` the **whole chain at once** — one equation per stage — rather than three disconnected regressions. That matters: because the model knows the full graph, it can propagate an intervention on onboarding *through* engagement and activation to conversion, giving genuine path-specific effects instead of naive stage-by-stage correlations. The `effects_summary()` below lists every structural coefficient (the a's and b's from Step 3); Step 5 then combines them into total, direct, and indirect effects with uncertainty.

Throughout, uncertainty is reported as a **credible interval** — the Bayesian analogue of a confidence interval, the range that contains the parameter with the stated posterior probability; note the `effects_summary()` table below shows a **94% HDI (highest-density interval)** — the narrowest credible interval containing 94% of the posterior — (`hdi_3%`/`hdi_97%`) while Step 5's print reports a **90%** interval.

A **prior** is the belief we hold about a coefficient *before* seeing data; after fitting, the **posterior** is that belief updated by the data — every effect below is read off this posterior, so it comes with a spread, not just a point.

Reading the sampler's health check (printed right after the fit). Three numbers say whether the MCMC sampler — the algorithm that draws samples from the posterior — actually converged: **R-hat** compares the variance within each chain to the variance across chains (1.00 is perfect, **$\leq$ 1.01 is the usual pass bar**); **ESS** (effective sample size) is how many *independent* draws our autocorrelated chains are worth (a few hundred is ample for a posterior mean or interval); **divergences** are steps where the sampler broke down and distrusts that region — **you want 0**. Under this notebook's FAST teaching profile the chains are deliberately short, so R-hat can drift to ~1.02 and PyMC may print a "problems during sampling" notice — a benign small-draw artifact that clears in a FULL run — as here: this committed FULL fit reports R-hat at/near 1.00, ESS in the thousands, and 0 divergences; the fresh-seed replication in Step 5 is the real backstop.

The fitted model, in symbols — exactly the three structural equations of step 2, refit with a coefficient posterior on every arrow, *plus* one extra edge (`c_on`, a direct $O \to$ converted arrow) we include deliberately even though we suspect it is zero, so "how much is mediated?" is *discovered* (its posterior should sit at ≈ 0) rather than assumed by omission. Each coefficient gets a weakly-informative normal prior — printed, written out in full, and stress-tested immediately after the fit — and `effect(path)` returns the **product of the posterior coefficient draws along the path** — e.g. the long path's effect is the posterior of $e_{on} \cdot a_{eng} \cdot c_{act}$, uncertainty included.

```
mediation model convergence: max r-hat 1.000 - min ESS 3737 - divergences 0

        mean     sd  hdi_3%  hdi_97%
name
e_on   2.153  0.045   2.074    2.241
e_ch   1.132  0.044   1.053    1.222
a_eng  1.617  0.018   1.585    1.652
a_ch   0.592  0.051   0.491    0.681
c_act  0.901  0.022   0.858    0.941
c_eng  0.336  0.045   0.251    0.420
c_on   0.060  0.055  -0.044    0.162
```

The model we actually fit — likelihood, priors, and whether they matter

Before reading a single effect, write the model down. `pathmc` compiled the spec into three Gaussian likelihoods — one per stage — each with an intercept the DGP doesn't have (harmless; its posterior sits at ≈ 0) and a weakly-informative prior on every coefficient:

$$
\begin{aligned}
\text{engagement}_i &\sim \mathcal{N}(\beta_0^{(1)} + e_{on}\, O_i + e_{ch}\, Q_i,\ \sigma_1^2),\\
\text{activated}_i &\sim \mathcal{N}(\beta_0^{(2)} + a_{eng}\, \text{engagement}_i + a_{ch}\, Q_i,\ \sigma_2^2),\\
\text{converted}_i &\sim \mathcal{N}(\beta_0^{(3)} + c_{act}\, \text{activated}_i + c_{eng}\, \text{engagement}_i + c_{on}\, O_i,\ \sigma_3^2),\\
\beta_\bullet &\sim \mathcal{N}(0,\ 10^2), \qquad \sigma_\bullet \sim \text{HalfNormal}(1).
\end{aligned}
$$

Why these priors are safe here. Every variable lives on a roughly unit scale ($O, Q \in [0,1]$, the stages are small multiples of them), the planted coefficients are all below 2.5 and the noise scales 0.3–0.4 — so $\mathcal{N}(0, 10^2)$ is diffuse relative to any plausible effect and HalfNormal(1) generous for the noise, and with $n = 1000$ the likelihood should dominate. "Should" is checkable: the cell below **doubles every prior scale and refits**. If the posterior means move by more than Monte-Carlo error, the priors were doing the talking; if nothing moves, the data set these numbers. We also finally read the Step-4 coefficient table **against the planted truth**, arrow by arrow — the path products of Step 5 are only as good as the arrows they multiply.

```
Priors:
  beta_engagement: Normal(mu=0, sigma=10)
  sigma_engagement: HalfNormal(sigma=1)
  beta_activated: Normal(mu=0, sigma=10)
  sigma_activated: HalfNormal(sigma=1)
  beta_converted: Normal(mu=0, sigma=10)
  sigma_converted: HalfNormal(sigma=1)
```

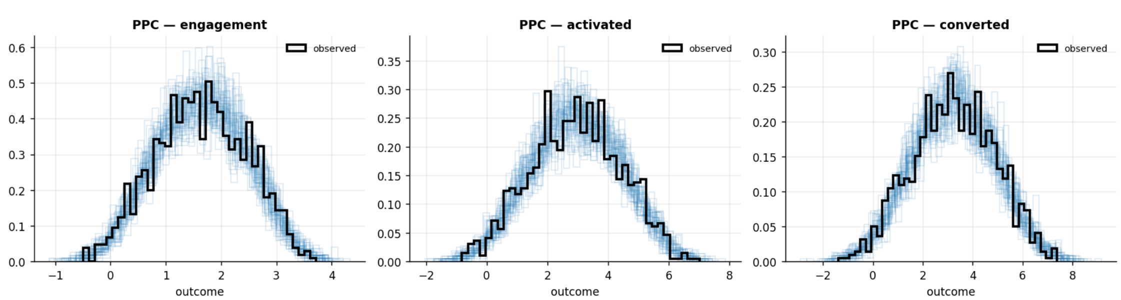
```
       posterior mean  posterior sd  planted truth    |z|
name
e_on            2.153         0.045           2.20  1.059
e_ch            1.132         0.044           1.10  0.713
a_eng           1.617         0.018           1.60  0.948
a_ch            0.592         0.051           0.60  0.163
c_act           0.901         0.022           0.90  0.034
c_eng           0.336         0.045           0.35  0.300
c_on            0.060         0.055           0.00  1.093
Every coefficient's posterior mean sits within 2 posterior sd of its planted
value (worst |z| = 1.1) -- the individual arrows are recovered, so the path
products in Step 5 rest on sound pieces.
```

```
Doubling every prior scale (Normal(0,10) -> Normal(0,20)) moves the largest
posterior mean by 0.001 (0.04 posterior sd) -- the data, not the priors, set
these numbers.
```

Model criticism — a posterior predictive check per stage equation

Convergence says the sampler explored the posterior; it says nothing about whether the **model can generate the data**. A mediation model is *three* likelihoods chained together, and the failure mode is multiplicative: a misfit in any one stage equation contaminates **every path product built through it** — an over-dispersed engagement equation, say, corrupts both the engagement lever *and* onboarding's total, because both ride the same fitted edges. So the PPC discipline from the rest of the cookbook ("check the model can reproduce the data before reading effects off it") applies here **once per equation**: we draw replicated datasets from the fitted chain and overlay each stage's replicated distribution on the observed one, with a quantitative read-out — the share of real observations inside each user's central 90% replicate band.

```
Share of observed values inside the replicates' per-user 5-95% band (~90% if
calibrated):  engagement: 88%  *  activated: 89%  *  converted: 90%
No gross misfit in any stage equation -- safe to read path effects off this
model.
```



Real-data check + the naive-regression trap. Before trusting the split: (a) do the funnel **DAG**'s (directed acyclic graph — the arrows-only causal diagram) own testable implications survive on this data — each reported below as a **partial correlation** (the correlation left between two variables after removing the conditioning set), and (b) what would a *naive* controlled regression — onboarding **plus** the mediators, all in one OLS — report for onboarding? The path queries above need neither trick; the naive regression, as the numbers show, misreads onboarding's total effect.

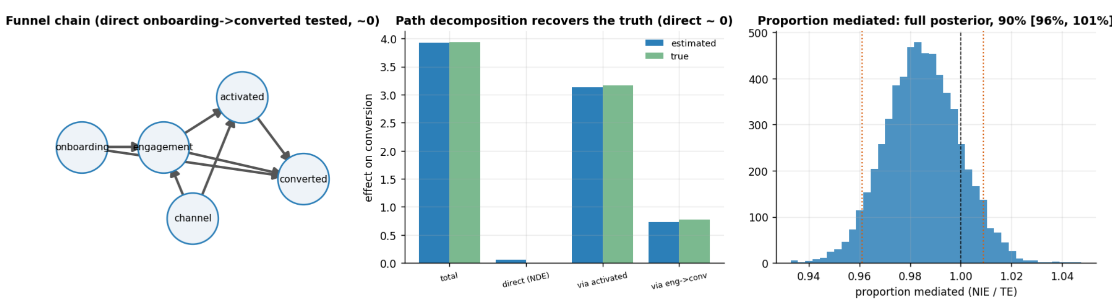
```
Testable implications of the funnel DAG: 3 checked -> 1 violated; P(>=1 by
chance at alpha=0.05) = 0.14 (graph survives falsification).
  the flagged independence: channel_quality _||_ onboarding_score | {}  (partial
corr +0.087, p = 0.006) - in this simulation every implied independence holds by
construction, so this is a knowable false positive, exactly what the Binomial
budget above prices in.
Naive OLS  converted ~ onboarding + engagement + activated : onboarding coef =
+0.059  (vs TRUE total 3.94).
Putting the mediators in the regression collapses onboarding's coefficient
toward ~0 (it now reads only the leftover DIRECT effect, not the total). That is
exactly why a naive controlled regression MISreads onboarding's real leverage --
and why the path decomposition above, not OLS, is what you need.
```

5 · Validate — recover the total, the paths, the DIRECT effect, and the proportion mediated

We query four things and check each against the planted truth: the **total** effect of onboarding on conversion; the effect flowing through each **path**; the **direct** (natural direct) effect — which the model is now free to find, and which should come out $\approx$ **0** if onboarding really works only through the funnel; and the **proportion mediated** (indirect $\div$ total). A proportion near 100% *with a direct effect whose interval includes 0* is the honest version of "onboarding works almost entirely by feeding the rest of the funnel" — a very different investment story from a direct effect.

Recovery on one dataset is necessary, not sufficient — so Step 5 closes with two harder checks: a **replication across fresh simulations** (is the recovery calibrated, or a lucky seed?) and a **fail $\rightarrow$ fix** refit that deliberately drops the mediators' common driver to show what unmeasured mediator confounding actually does to the split.

```
TOTAL onboarding -> converted           3.92  (true 3.94)  [90% 3.76, 4.08]
  DIRECT (NDE) onboarding -> converted  0.06  (true 0.00)  [90% -0.03, 0.15]
  via engagement->activated->conv       3.14  (true 3.17)  [90% 2.96, 3.31]
  via engagement->converted             0.72  (true 0.77)  [90% 0.56, 0.89]
Proportion mediated (indirect / total): 98%   [90% 96%, 101%]
The DIRECT effect's 90% interval includes 0, so the ~98% proportion mediated is
DISCOVERED - the model was free to find a direct onboarding->conversion path and
the data say there isn't one.
```



How to read this. *Left* is the causal graph we fit — including a **direct** onboarding→conversion arrow, which we deliberately added so the model could find a direct effect if one existed. *Middle* — the estimated total and each path effect (blue) sit on top of the planted truths (green), and the **direct (NDE) bar is ≈ 0** (its interval includes zero): onboarding works almost entirely *through* the funnel, and that near-100%-mediated result is now **discovered, not assumed**. *Right* — the proportion mediated as a **full posterior** with its 90% interval, because "~100% mediated" is a headline number people will quote and it should carry uncertainty like everything else (the dashed line at 1.0 is "fully mediated"; draws land on either side of it because both NIE and TE are noisy). This is the decomposition a naive "regress conversion on each stage" analysis cannot give you.

Not a lucky seed — replication on fresh simulated worlds

Everything above is one dataset from one seed. A method that recovers the truth once might be benefiting from friendly noise, and its intervals might be too tight (they'd then miss the truth on other draws more than 10% of the time). So we redraw the world from the same DGP several times, refit with deliberately short chains (posterior *means and intervals* are all we grade), and check two things per quantity: **bias** (do the estimates center on the truth?) and **90%-interval coverage** (do the intervals contain it about 90% of the time?). Because the four path quantities are all read off the *same* fit, they do not fail independently: an unlucky dataset tends to shift them together, so any misses cluster on a seed rather than scattering one-by-one across the table.

```
planted truth:   total = 3.94   direct = 0.00   via_act = 3.17   via_eng = 0.77

 seed   total    direct   via_act   via_eng
  100 3.93 ok   0.02 ok   3.26 ok   0.65 ok
  101 4.02 ok  -0.03 ok   3.34 ok   0.71 ok
  102 3.84 ok  -0.00 ok   3.11 ok   0.73 ok
  103 3.90 ok   0.01 ok   3.22 ok   0.66 ok
  104 3.97 ok  -0.01 ok   3.19 ok   0.78 ok
  105 4.10 ok 0.11 MISS 3.40 MISS 0.59 MISS
  106 4.03 ok  -0.04 ok   3.19 ok   0.88 ok
  107 3.94 ok  -0.05 ok   3.11 ok   0.89 ok
(cell = posterior mean + 90%-interval coverage of the planted truth)

4 quantities x 8 fresh simulations -> 32 90% intervals, 29 covered (91%; ~90% is
calibrated).  Mean bias: total +0.03  direct +0.00  via_act +0.06  via_eng
-0.03.
Recovery is consistent across seeds -- Step 5's numbers are calibrated, not a
lucky draw.
```

Fail → fix: drop the mediators' common driver and watch the split break

Step 3 warned that with two ordered mediators, an **unmeasured confounder between the mediators** poisons the path decomposition — and this DGP hands us the perfect prop: `channel_quality` feeds *both* engagement and activation. So far we have always adjusted for it. Now we refit the same chain **with every `channel_quality` term deleted**, exactly as if the growth team had never logged acquisition channel — the single most realistic way this analysis goes wrong in production.

What should happen, mechanically: with Q gone, the engagement→activated edge **absorbs Q's effect on activation** (users on good channels are both more engaged and more activated, and the model can only route that association through `a_eng`), so `a_eng` overshoots its planted 1.6 — and every path product built on that edge (the long path, and the graph-computed total that sums the paths) inherits the bias. Meanwhile a plain OLS of conversion on onboarding *alone* stays honest, because onboarding itself is exogenous here (nothing feeds into it) — which is precisely the opening cell's claim, now with numbers: **unmeasured mediator confounding breaks the split (and everything computed from it) even when the raw total effect of the treatment is fine.** And the DAG falsification test from Step 4 — which on the correct graph only ever produced a priced-in false positive — should now catch a **true** violation. Read the printed output below against each prediction.

```
quantity  truth  with Q  with Q covers  no Q  no Q covers
   total   3.94    3.92           True  4.29        False
  direct   0.00    0.06           True  0.06         True
 via_act   3.17    3.14           True  3.47        False
 via_eng   0.77    0.72           True  0.76         True

With channel_quality dropped, eng->act absorbs Q's effect: a_eng = 1.71 vs
planted 1.60 -- and 2 of 4 'no Q' 90% intervals now miss the truth (vs 0 of 4
with Q).
Plain OLS converted ~ onboarding alone: 4.13 (SE 0.13) vs true 3.94 --
consistent with the truth (1.5 SE away). The split is what breaks; the raw total
was never in danger.

test_implications on the no-Q graph: 1 test(s) -> 1 violated:
  activated _||_ onboarding_score | {engagement} FAILS (partial corr -0.184, p =
5.1e-09) -- the unmeasured mediator confounder leaves a testable footprint; the
Step-4 falsification cell is not decoration, it catches exactly this
misspecification.
```

6 · Decide, in euros — a *rankable* leverage metric, its intervals, and its fragility

Three moves, and each corrects a trap.

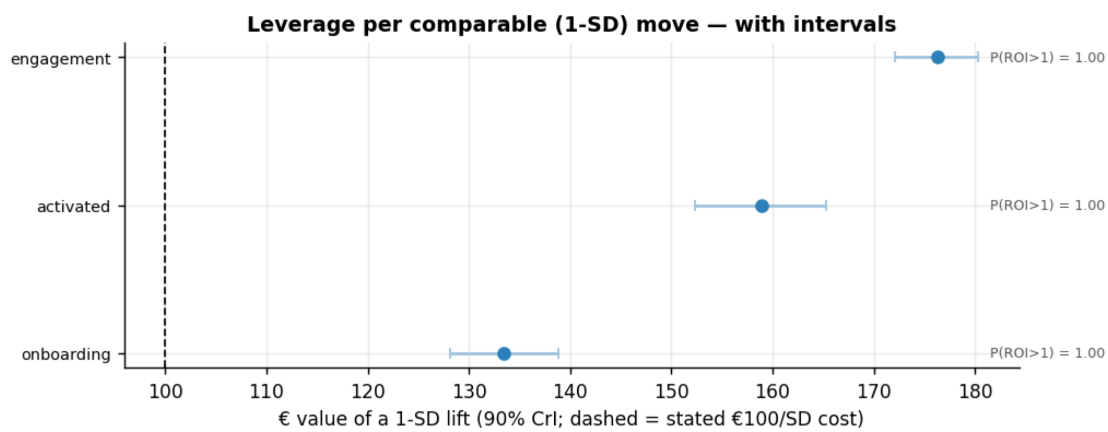
(1) Rank the levers — but not by "€ per one-unit change." A one-unit move means something completely different per stage (one unit of `onboarding_score` is a large fraction of its whole range; one unit of `activated` is a fraction of a standard deviation), so a €/unit ranking is not a rankable quantity — it *reverses* when you switch to comparable **per-standard-deviation** moves. The decision-relevant metric is **ROI = (€ value of a 1-SD lift) ÷ (€ cost to produce that 1-SD lift)** — and every € number below carries its **90% credible interval** (table + forest plot), because a ranking whose intervals overlap is a preference, not a decision.

(2) Turn the stated cost into a break-even rule. "Equal €100 per SD for every stage" is an illustration, not a fact — real per-stage costs are what the promise of a *budget split* hinges on. So instead of betting on one cost number, we report the **posterior break-even cost ratio** for each pair of levers: lever A beats lever B on ROI exactly when $\text{cost}_A/\text{cost}_B < \text{value}_A/\text{value}_B$, and the right-hand side is a posterior we already have. Hand the rule to finance: they measure the cost ratio, the posterior says who wins — with uncertainty attached.

(3) Stress-test the split — coherently. The decomposition rests on *no unmeasured mediator–outcome confounding* (SI-2). A confounder of engagement→conversion biases the fitted edge — but that same edge carries **both** the engagement lever **and** onboarding's total effect, so an honest sweep must shrink them *together* (a common mistake is to bias one and freeze the other). We do that per posterior draw — so the sweep has **bands**, and the "flip point" is a posterior with an interval, not a single number.

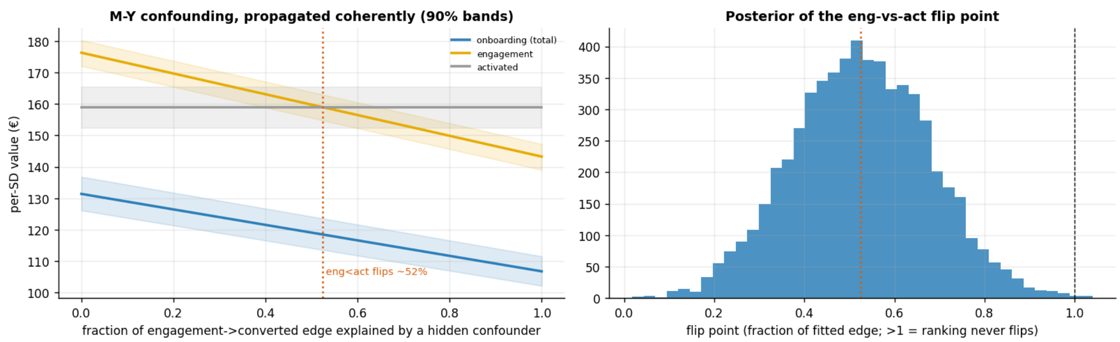
```
     lever  €/unit  €/SD €/SD 90% CrI  ROI (€/€)  P(ROI>1)
engagement     215   176   [172, 180]       1.76       1.0
 activated     108   159   [152, 165]       1.59       1.0
onboarding     470   133   [128, 139]       1.33       1.0

Per-UNIT vs per-SD DISAGREE: a 1-SD onboarding move is only 0.28 units (tiny)
while a 1-SD activated move is 1.47 units (large), so €/unit is not rankable --
use ROI on a stated cost. At equal €100/SD the order is engagement > activated >
onboarding; all three clear break-even at the stated cost.
Sampling-noise decisiveness of the eng>act gap: P(engagement > activated per SD)
= 1.00 (1.00 = no overlap, ordering certain up to sampling).
Break-even rules: onboarding beats engagement on ROI once a 1-SD onboarding lift
costs less than 0.76x a 1-SD engagement lift (90% [0.72, 0.79]); for activated
the break-even cost ratio vs engagement is 0.90x (90% [0.86, 0.95]).
```



How to read the forest plot. Every lever's €/SD value clears the stated €100 cost line with room to spare, and `P(ROI>1)` saturates — which is itself informative: **sampling uncertainty is not the binding risk here** (n=1000 and a linear SCM make these posteriors tight; the printed P(engagement > activated) quantifies exactly how decisive the equal-cost ordering is on this data). The intervals matter for the *break-even rules*: the posterior break-even ratios above are the numbers to hand to finance, and their 90% intervals are what an honest budget memo quotes. What the forest plot **cannot** show is assumption risk — the possibility that the engagement→conversion edge itself is partly a hidden confounder's work. That is the sweep below, and it is where the ranking actually becomes fragile.

```
The eng > act per-SD ordering flips once a hidden M-Y confounder explains ~52%
of the fitted engagement->converted edge (90% [27%, 77%]); P(no flip within the
whole edge) = 0.00.
Onboarding's TOTAL shrinks along the same sweep (it rides the same edge) but its
story -- a root investment whose value is almost fully mediated -- survives; the
fragile margin is engagement vs activated, and it should be quoted with its
interval, never as a single flip percentage.
```



How to read the sweep. The bands are the posterior 90% intervals *at each level of hypothetical confounding* — so the figure now separates the two kinds of uncertainty a decision-maker must not conflate: the (narrow) sampling bands, and the (wide) horizontal question of *how much of the fitted edge you believe*. The flip point itself is a posterior (right panel): quote its interval from the printed line, not the median alone. If domain knowledge says hidden drivers of both engagement and conversion (user intent, cohort quality) could plausibly account for a large share of that edge, the engagement-vs-activated ranking should be treated as undecided — while onboarding's near-fully-mediated total survives the same sweep with its story intact.

The one-paragraph decision

> **What you tell the CMO.** Onboarding's conversion value is essentially **all mediated** ($\approx$ 100%, with its 90% interval printed in Step 5): it is a *root* investment whose payoff lands two stages downstream, so judging it by direct/last-touch conversion attribution — as the naive regression in Step 4 effectively does — would price it at roughly zero and be exactly wrong. Per comparable (1-SD) move at equal cost, the ranking on this data is **engagement > activated > onboarding** (forest plot above; the printed P tells you how decisive the eng-vs-act gap is), **but that margin is the fragile one**: it flips once a hidden mediator–outcome confounder explains roughly half of the fitted engagement→conversion edge (interval printed above), whereas onboarding's total is not similarly at risk — its identification leans on the treatment side, which is exogenous here. The equal-cost assumption is disposable: by the break-even rule above, onboarding wins on ROI whenever a 1-SD onboarding lift costs less than about 0.8× a 1-SD engagement lift (interval printed above) — so the *budget split* question reduces to a cost measurement finance can actually do. The follow-up is an experiment that **randomizes an onboarding improvement and logs the mediators**: randomization

buys SI-1 by design, and measured mediators let you re-estimate the split with half the assumptions earned rather than assumed.

The machine-readable version, for the deck and the decision log:


```
{
  "true_total": 3.938,
  "est_total_mean": 3.919,
  "est_total_90": [
    3.763,
    4.078
  ],
  "nde_mean": 0.06,
  "prop_mediated_mean": 0.985,
  "prop_mediated_90": [
    0.961,
    1.009
  ],
  "value_per_sd_eur": {
    "onboarding": 133.433,
    "engagement": 176.303,
    "activated": 158.908
  },
  "ranking_per_sd_equal_cost": [
    "engagement",
    "activated",
    "onboarding"
  ],
  "p_engagement_beats_activated_per_sd": 1.0,
  "flip_fraction_median": 0.525,
  "flip_fraction_90": [
    0.275,
    0.773
  ],
  "breakeven_cost_ratio_onboarding_vs_engagement": {
    "median": 0.757,
    "cri90": [
      0.725,
      0.788
    ]
  },
  "breakeven_cost_ratio_activated_vs_engagement": {
    "median": 0.902,
    "cri90": [
      0.857,
      0.947
    ]
  },
  "replication_coverage_90": 0.906,
```

```
  "ppc_band_share": {
    "engagement": 0.885,
    "activated": 0.891,
    "converted": 0.895
  }
}
```

7 · Caveats

- **Natural effects lean on strong assumptions** — SI-1–SI-4 from Step 3, plus (with two ordered mediators) no unmeasured confounding *between* the mediators. Step 5 showed with numbers what the latter's failure does (the fail → fix refit), and Step 6 quantified *coherently* (shrinking both affected levers together) how much M–Y confounding would flip the leverage ranking; the point estimate alone hides that the engagement-vs-activation ordering is the fragile one.
- **Mediator measurement error attenuates the indirect effect.** If "engagement" is a noisy proxy for the behaviour that actually drives conversion (in product analytics it always is), the fitted mediator→outcome coefficient b is biased toward zero, so the NIE is *understated* and the missing share silently reappears as "direct" effect. A near-zero NDE like ours is therefore doubly reassuring; a *large* NDE on real data should always be interrogated for mediator noise first.
- **No treatment–mediator interaction is baked in.** NDE + NIE = TE holds exactly here because the SCM has no $X \times M$ term. If onboarding *changes the return to* engagement (plausible: better-onboarded users engage more productively), the clean two-way split no longer adds up and you need the four-way decomposition of VanderWeele (2014) — a model-specification caveat to check before porting this notebook to real data.
- **A €/unit leverage ranking is not a rankable quantity** when the stage metrics live on different scales — it reverses under per-SD units. Decide on **ROI against a stated cost-to-move** (or better, the break-even cost *ratio* with its posterior interval), not €/unit.
- **Don't control away the mechanism.** The goal is to *decompose* the path; adjusting for a mediator when you wanted the total effect removes the very thing you're pricing (nb 05).
- **Linear-SCM decomposition is exact only if the SCM is linear** — and linearity is an *economic* assumption, not just a statistical one: a linear model prices the 10th SD of engagement the same as the 1st, i.e. **no diminishing returns**, which is exactly where a CMO will push back. Read every €/SD figure as a *local, first-SD* number, and for strong nonlinearities lean on `pathmc`'s computed effects, not hand-multiplied coefficients.